Exercise Sheet 6

MATH 598 / 784 · High-Dimensional Probability

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Throughout, $C > 0$ denotes an absolute constant, and $w(T) = \mathbb{E}\sup_{t\in T}\langle g, t\rangle$ for a standard Gaussian vector g in the ambient space. All index sets are nonempty.

1. Gaussian contraction (Exercise 7.2.13)

Consider a bounded subset $T \subset \mathbb{R}^n$, and let $g_1, \dots, g_n$ be independent $N(0,1)$ random variables. Let $\phi_i : \mathbb{R} \to \mathbb{R}$ be contractions, i.e. Lipschitz functions with $\|\phi_i\|_{\mathrm{Lip}} \le 1$. Prove that

$$\mathbb{E}\sup_{t\in T}\sum_{i=1}^{n} g_i\phi_i(t_i) \le \mathbb{E}\sup_{t\in T}\sum_{i=1}^{n} g_i t_i.$$

Hint. Use Sudakov–Fernique's inequality.

2. Gordon's inequality (Exercise 7.2.14)

Let $(X_{ut})_{u\in U, t\in T}$ and $(Y_{ut})_{u\in U, t\in T}$ be centered Gaussian processes, where U and T are finite sets. Assume that

$$\begin{aligned}
\mathbb{E}X_{ut}^2 &= \mathbb{E}Y_{ut}^2 && \text{for all } u, t,\\
\mathbb{E}(X_{ut} - X_{us})^2 &\le \mathbb{E}(Y_{ut} - Y_{us})^2 && \text{for all } u, t, s,\\
\mathbb{E}(X_{ut} - X_{vs})^2 &\ge \mathbb{E}(Y_{ut} - Y_{vs})^2 && \text{for all } u \ne v \text{ and all } t, s.
\end{aligned}$$

Prove that, for every $\tau \in \mathbb{R}$,

$$\mathbb{P}\left\{\min_{u\in U}\max_{t\in T} X_{ut} \ge \tau\right\} \le \mathbb{P}\left\{\min_{u\in U}\max_{t\in T} Y_{ut} \ge \tau\right\}.$$

Deduce that

$$\mathbb{E}\min_{u\in U}\max_{t\in T} X_{ut} \le \mathbb{E}\min_{u\in U}\max_{t\in T} Y_{ut}.$$

For continuous processes on compact index sets, explain how to extend the expectation comparison by passing to finite nets, assuming that both processes have integrable suprema of their absolute values.

Hint. Apply the Gaussian interpolation lemma to

$$f(x) = \prod_i \left[1 - \prod_j h(x_{ij})\right],$$

where h is smooth, nonincreasing, takes values in $[0,1]$, and approximates $\mathbf{1}_{\{x<\tau\}}$. Check the signs of the mixed second derivatives within a row and between different rows. For the expectation, integrate the tail comparisons on both sides of zero.

3. From Gaussian width to a Johnson–Lindenstrauss embedding

Let G be an $m \times n$ matrix with independent $N(0,1)$ entries, let $S \subseteq S^{n-1}$ be compact, and set $a_m = \mathbb{E}\|h\|_2$, where $h \sim N(0, I_m)$.

1. Use Sudakov–Fernique's inequality and Problem 2 to prove

$$\mathbb{E} \sup_{x \in S} \|Gx\|_2 \le a_m + w(S), \qquad \mathbb{E} \inf_{x \in S} \|Gx\|_2 \ge a_m - w(S).$$

2. Show that the supremum and infimum in part 1 are 1-Lipschitz functions of G for the Frobenius norm. Deduce that, for every $r \ge 0$, with probability at least $1 - 2e^{-r^2/2}$,

$$\sup_{x \in S} \left| \|Gx\|_2 - \sqrt{m} \right| \le w(S) + r + 1.$$

   You may use $0 \le \sqrt{m} - a_m \le 1$, as follows from Problem 3 of Sheet 5 with $A = I_m$.

3. For a set $T \subseteq \mathbb{R}^n$ containing at least two points, define its set of normalized secants by

$$S(T) = \overline{\left\{ \frac{x-y}{\|x-y\|_2} : x, y \in T,\ x \ne y \right\}}.$$

   Given $\varepsilon, \rho \in (0,1)$, prove that, with probability at least $1 - \rho$, the map $\Phi = G/\sqrt{m}$ satisfies

$$(1-\varepsilon)\|x-y\|_2 \le \|\Phi x - \Phi y\|_2 \le (1+\varepsilon)\|x-y\|_2 \qquad \text{for all } x, y \in T,$$

   provided

$$m \ge C\varepsilon^{-2} \left(w(S(T))^2 + \log \frac{2}{\rho} \right).$$

   When T has $N \ge 2$ points, bound $w(S(T))$ to recover the classical dimension $m \ge C\varepsilon^{-2} \log(N/\rho)$.

4. **Why normalize the secants?** Show that if T has nonempty interior, any linear map satisfying the lower bound in part 3 must have $m \ge n$. For $n \ge 2$, consider the thin ellipsoid

$$E_n = \left\{ x \in \mathbb{R}^n : x_1^2 + n \sum_{i=2}^{n} x_i^2 \le 1 \right\}.$$

   Check that $w(E_n)^2 / \operatorname{diam}(E_n)^2 \le 1/2$. Explain why this ratio alone cannot control the dimension needed to preserve every distance with a multiplicative error.

Hint for part 1. For independent $g \sim N(0, I_n)$, $h \sim N(0, I_m)$, and $z \sim N(0,1)$, compare the processes

$$\langle Gx, y \rangle, \qquad \langle g, x \rangle + \langle h, y \rangle, \qquad \langle Gx, y \rangle + z, \qquad (x, y) \in S \times S^{m-1}.$$

The scalar z is shared by every index pair and makes the variances equal for Gordon's inequality. Approximate the compact index sets by finite nets.

4. Gaussian Chevet and stable rank (Exercise 8.7.4)

Let $T \subseteq \mathbb{R}^n$ and $S \subseteq \mathbb{R}^m$ be compact sets symmetric about the origin, and write $\mathrm{rad}(T) = \sup_{x \in T} \|x\|_2$. Let G be an $m \times n$ standard Gaussian matrix.

1. Prove the Gaussian Chevet inequality with constant 1:

$$\mathbb{E} \sup_{x \in T,\, y \in S} \langle Gx, y \rangle \le \mathrm{rad}(S) w(T) + \mathrm{rad}(T) w(S).$$

2. Let $A \in \mathbb{R}^{p \times m}$ and $B \in \mathbb{R}^{n \times q}$ be deterministic matrices. Apply part 1 to suitable ellipsoids to show that

$$\mathbb{E}\|AGB\|_{\mathrm{op}} \le \|A\|_{\mathrm{op}} \|B\|_F + \|A\|_F \|B\|_{\mathrm{op}}.$$

   For nonzero A, B, rewrite the bound as

$$\mathbb{E}\|AGB\|_{\mathrm{op}} \le \|A\|_{\mathrm{op}} \|B\|_{\mathrm{op}} \left(\sqrt{r(A)} + \sqrt{r(B)} \right), \qquad r(M) = \frac{\|M\|_F^2}{\|M\|_{\mathrm{op}}^2}.$$

   Check that $A = I_m$, $B = I_n$ recovers the bound $\mathbb{E}\|G\|_{\mathrm{op}} \le \sqrt{m} + \sqrt{n}$.

Hint for part 1. Take independent standard Gaussian $g \in \mathbb{R}^n$, $h \in \mathbb{R}^m$, and $z \in \mathbb{R}$, independent also of G, and compare

$$X_{xy} = \langle Gx, y \rangle + z \|x\|_2 \|y\|_2, \qquad Y_{xy} = \|y\|_2 \langle g, x \rangle + \|x\|_2 \langle h, y \rangle.$$

Their variances agree. For two index pairs, their covariance difference factors into two nonnegative terms. Use comparison and Jensen's inequality to remove the added scalar Gaussian. Where is symmetry used?

Problems 1 and 2 are adapted from Exercises 7.2.13–7.2.14 of Roman Vershynin, *High-Dimensional Probability*, first edition. Problem 2 asks for all real thresholds. Problem 3 combines the comparison argument behind Exercise 7.3.4 with the normalized-secant construction in Section 9.3.1; its last part develops Exercise 9.3.3. It requires no chaining or general matrix deviation theorem. Problem 4 specializes Exercise 8.7.4 to symmetric sets and adds a stable-rank application. The author provides the first-edition PDF free of charge. Numbering differs in the second edition.